Software Engineering Lab - Lab 6

Code and Branch Coverage Analysis

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Assigned program - Income Tax Calculator

1. Objective

Measure statement and branch coverage of the assigned income tax program using pytest and Coverage.py. Additional tests cover valid inputs, invalid inputs and alternate decision outcomes. The provided application code was not modified.

2. Test Case Design

Function	Conditions covered
taxable_income()	New and old regimes; investment deduction limits; negative gross income; unknown regime; zero taxable income.
tax_due()	Age groups below 60, 60-79 and 80+; exemptions, tax slab boundaries and negative amounts.
refund_status()	Refund, delayed refund, balance due, balance with penalty and settled cases.

3. Statement Coverage

Test file: CodeCoverageTest_income_tax.py

The statement coverage suite contains 14 test cases. All executable statements were exercised.

```
Terminal

$ python -m coverage run --source=income_tax -m pytest -q CodeCoverageTest_income_tax.py
..............                                                           [100%]
14 passed in 0.61s
$ python -m coverage report -m
Name            Stmts   Miss  Cover   Missing
---------------------------------------------
income_tax.py      39      0   100%
---------------------------------------------
TOTAL              39      0   100%
```

Figure 1. Statement coverage result: 14 tests passed; 39 of 39 statements covered.

4. Branch Coverage

Test file: BranchCoverageTest_income_tax.py

The branch coverage suite checks both outcomes of conditional statements, including regime selection, investment caps, age thresholds, tax slab boundaries and refund decisions.

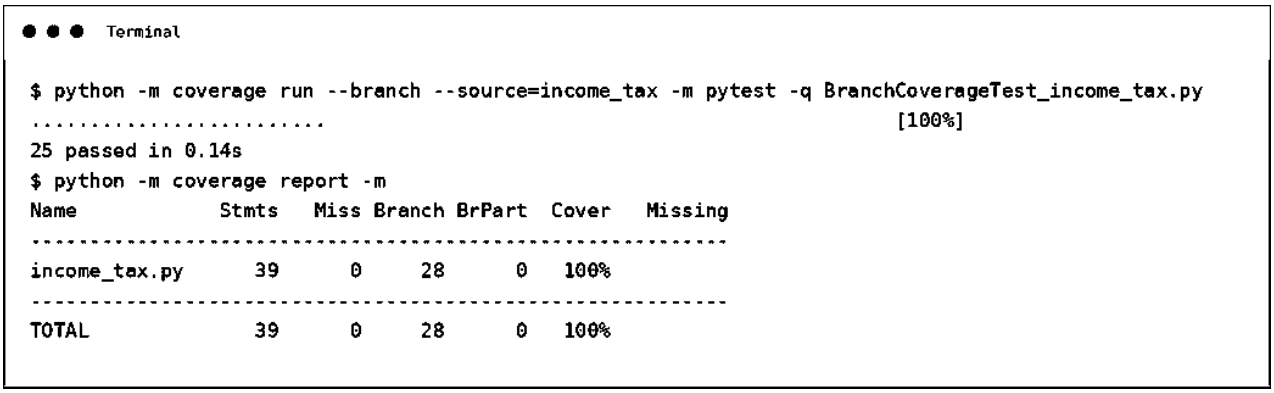

Figure 2. Branch coverage result: 25 tests passed; 28 of 28 branches covered.

5. Coverage Summary

Coverage type	Tests passed	Statements missed	Branches missed	Coverage
Statement	14	0 of 39	-	100%
Branch	25	0 of 39	0 of 28	100%

Both statement and branch coverage exceed the minimum requirement of 90%. The two coverage stages are recorded as separate Git commits.

6. Result

All 39 statements and all 28 branches of the assigned module were covered without changing the application source.